

Arian Niakan

Game Design Student at BUG

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EDUCATION

Master's degree

BAU Bahçeşehir University

- Game Design

Oct 2025 - Present

Istanbul-Turkey

Bachelor's degree

UT University of Tehran

- Electrical Engineering (Digital Systems)

Sep 2019 - Sep 2024

Tehran-Iran

RESEARCH and WORK EXPERIENCE

🎮 Gameln · Volunteer

Jul 2021 - Dec 2021

- Game Designer

- Designed an online game for a competition event, simulating the space of industrial work, where players would compete for resources by buying raw materials, creating products, and etc.
- Learned the iterative process of game design.

NBIC University of Tehran's NBIC Research Institute · Research

Sep 2023 - Sep 2025

- Time Perception with Multi-Sensory stimuli (audio, visual, and haptic) research assistant.
 - Handling task design, data collection device(EEG and Behavioural) and processing/modeling.
 - Working with Prof. Simon Grondin and Prof. Mohammad-Reza A. Dehaqani's team on several forms of the interval discrimination task and the equisection task.
 - Created a game controller like device that can handle Haptic, Audio, and visual stimuli using a raspberry pi that can run many necessary task's relating to time perception using Python code(or C++).

NBIC University of Tehran's NBIC Research Institute · Research/Educational

July 2024 - Sep 2024

- Summer Internship Supervisor.

- Provided planning, supervision, and support for three interns working on the continuation of the time perception project at the lab of Prof. Mohammad-Reza A. Dehaqani.

🌐 Quera: Online Programming Skill Assessment · Educational

June 2024 - Sep 2024

- Data Science Team Member.

- Writing Natural Language Processing (NLP) course's textbook, instructions, exercises, and course projects.
- Writing and testing contest questions using public or crawled data for general data science contests.

UT University of Tehran · Research/Industrial

July 2023 - May 2024

- Hardware Security and Encryption researcher.

- Working with Prof. Zainalabedin Navabi and Prof. Saeed Safari and their team on the Encryption and Cryptography of a custom SOC.

UT University of Tehran · Research

Oct 2022 - Aug 2023

- Field-Programmable Gate Arrays (FPGA) Engineer

- Worked with Prof. Zainalabedin Navabi and his team on the FPGA implementation of a custom RISC-V processor (called SAYAC), for prototyping and side-channel security testing purposes.

CROUSE PJS Co. · Internship

Aug 2022 - Sep 2022

- Hardware design engineering intern

- Worked as part of the research and development team of CROUSE PJS Co. motor company to develop an airbag micro-controller tester system by emulation an acceleration sensor using a Zynq SOC FPGA.

TEACHING ASSISTANCE EXPERIENCE

UT University of Tehran · TA

- Prof. Zainalabedin Navabi's Teaching Assistant

- Worked as a teaching assistant on the Digital Systems Design and Logic Gates course project.
- Prof. Saeed Safari's Lab Teaching Assistant
 - Worked as a teaching assistant at the Computer Architecture Lab.

SKILLS

Game Development & Programming

- Unity (C# scripting) & Godot (GDScript/C#)
- Gameplay Prototyping, Rapid Iteration, Game Feel Tuning
- Python (TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, Django)
- C/C++ (OOP, CUDA) for performance-sensitive systems and Hardware design
- Matlab; signal and EEG data processing; probability & statistics; AI modeling
- Gameplay ideation and iterative design for competition projects

Computing Hardware & Embedded Systems

- Verilog/SystemVerilog, VHDL, SystemC
- FPGA design on Altera and Xilinx Zynq SoC; ModelSim, Vivado, Quartus
- Embedded programming with Raspberry Pi and custom peripherals
- System-level, RTL, and gate-level digital design methodologies

Game Development Demos

- [Unity Mini-Games Collection](#) — Simple mechanics using free asset packs for visuals to showcase C# scripting, gameplay logic, and scene setup.

Cool Research Projects

- [Classification of Brain tumor types from MRI images using Convolutional Neural Network \(AI course project\)](#)
- [Re-simulation of a two-compartment hippocampal pyramidal cell model from this paper](#), and bifurcation diagram generation/analysis using XPP Auto. [Slides are available here.](#)
- [An Interactive view on famous Neural Models and Experiments](#) (like the Hodgkin-Huxley model and Voltage Clamp Experiment) the user can use sliding widgets to change the variables of different models.
- Simulations of the Models like the LIF model, fitzhugh-nagumo model, and ferguson-campbellare models. [here](#) and [here](#)
- [an analysis of rewards affect on attention \(from Neuromatch Academy's summer school\)](#)

COURSES and CERTIFICATES

-  [University of Tehran](#) · *Core Coursework*: Advanced Programming, Data Structures & Algorithms, Artificial Intelligence, SystemC OOP Modeling, Computer Architecture, Digital/Embedded/System-Level Design, Signals & Systems, Linear Control.
-  [Coursera](#): [Programming Fundamentals \(Duke\)](#); [Python Basics \(Michigan\)](#).
-  [Neuromatch Academy's Computational Neuroscience](#) (Jul 2023–Aug 2023): Team project modelling reward-driven attention; covered GLMs, dimensionality reduction, deep learning, neural dynamics, stochastic control, RL. [Slides](#) ▪ [Colab](#)
-  [Udemy](#): Complete Neural Signal Processing and Analysis (audited).

AWARDS

-  *Silver medal of the Domestic Olympiad on Astronomy and Astrophysics*
 - Iran's National Elites Foundation

LANGUAGE SKILLS

Persian

- Native

English

- Fluent
 - TOEFL Score: 114
 - Reading: 30
 - Listening: 30
 - Speaking: 28
 - Writing: 26

French

- Beginner